

Patent Specification: Constant-Time In-Memory Computing Architecture for Side-Channel Leakage Mitigation

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Abstract

This invention discloses a hardware-enforced constant-time key-management and current-balancing crossbar architecture for In-Memory Computing (IMC) systems. By equalizing dynamic charge draw (I_{total}) across read operations regardless of stored bit representations, the system eliminates power-side-channel leakage during KV-cache token retrieval in transformer inference workloads.

1 Background of Invention

Prior art implementations rely on dynamic software masking or delay insertion, introducing severe performance overheads ($> 30\%$). Current physical crossbars suffer from parasitic capacitive coupling, leaking secret key information during vector-matrix operations.

2 Detailed Description & Mathematical Model

The proposed system enforces a constant current draw according to Equation 1:

$$I_{rail}(t) = I_{data}(t) + I_{dummy}(t) = K \quad (\text{Constant}) \quad (1)$$

Where $I_{dummy}(t)$ is dynamically generated by a complementary crossbar differential element.

3 Patent Claims

1. An In-Memory Computing (IMC) circuit comprising: a primary crossbar memory array, a dummy-charge balancing array, and a dynamic differential current regulator enforcing dynamic power invariance across distinct data read cycles.
2. The circuit of claim 1, wherein bit-line charge equalizers operate within a clock domain boundary without software cycle delays.
2. The compute-in-memory architecture of claim 1, further comprising a dynamic dummy charge-balancing circuit coupled to each bitline pair, wherein said dynamic dummy charge-balancing circuit injects a complementary differential charge equal to $Q_{dummy} = Q_{max} - Q_{actual}$ during each computational accumulation cycle to maintain constant total array current draw.
3. The circuit of claim 2, wherein the dynamic dummy charge-balancing circuit comprises an array of programmable switch-capacitor banks controlled by an inverted weight-activation decode matrix operating in parallel with wordline activation.

4. The compute-in-memory architecture of claim 1, wherein the hardware-enforced constant-time execution pipeline guarantees uniform memory-access latency and Hamming weight power dissipation across all dynamic weight loading operations within the Key-Value cache.
5. The circuit of claim 2, further comprising a localized sense-amplifier power-rail regulator configured to damp differential voltage spikes induced by dummy charge injection within less than 150 picoseconds.